Firmware Analysis Report

Project: Firmware_120626.zip
Architecture: Unknown
File Size: 280180191 bytes
SHA256: a6c41ec01b31ec93ed4894427532c8cf2d67c3c23cc0310740b8e2fac084ca48

Executive Summary

This report was automatically generated by the FAWS system. It contains the results of a multi-stage static, dynamic, and symbolic firmware analysis.

Analysis Stages

Tool: STRINGS

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/strings.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip" --project "29" --run-id 0

```
Starting Strings Analysis on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow
simulator\backend\data\uploads\29_Firmware_120626.zip... Extracted 3206357 strings. Found 86824
finding(s). [!] FILE_PATH: Firmware_120626/GE.zipPK [!] FILE_PATH: GE/ux [!] FILE_PATH:
GE/Firmware File _ Oct 2023_72G.zipux [!] FILE_PATH: &/.,)M(hif [!] FILE_PATH: /HD<H [!]
FILE_PATH: 2/14D [!] FILE_PATH: 0/pz [!] FILE_PATH: /s7L [!] FILE_PATH: C$/q5 [!] FILE_PATH:
+S/5 Scan complete.
```

Tool: BINWALK

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/binwalk.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip" --project "29" --run-id 0

```
Starting Python Binwalk Engine on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow
simulator\backend\data\uploads\29_Firmware_120626.zip... Found 132 finding(s). [!] Offset 0x0:
Zip archive data [!] Offset 0x34: Zip archive data [!] Offset 0x75: Zip archive data [!] Offset
0x2da741: JPEG image data [!] Offset 0x33f031: gzip compressed data [!] Offset 0x4c65cf: JPEG
image data [!] Offset 0x50281e: JPEG image data [!] Offset 0x5ac47f: gzip compressed data [!]
Offset 0x7ba78c: JPEG image data [!] Offset 0x7c2572: Zip archive data [!] Offset 0x7dd811: Zip
archive data [!] Offset 0x7e7a6b: Zip archive data [!] Offset 0x838562: Zip archive data [!]
Offset 0x8414f7: Zip archive data [!] Offset 0x8966aa: Zip archive data [!] Offset 0x8f6661:
Zip archive data [!] Offset 0x9228c1: Zip archive data [!] Offset 0x931185: Zip archive data
[!] Offset 0x94ba14: Zip archive data [!] Offset 0x955efb: Zip archive data [!] Offset
0x9bbe7d: Zip archive data [!] Offset 0xa24d57: Zip archive data [!] Offset 0xa2d1ac: Zip
```

```
archive data [!] Offset 0xa5a994: Zip archive data [!] Offset 0xa742fe: Zip archive data [!]
Offset 0xa7d37d: Zip archive data [!] Offset 0xaa9979: JPEG image data [!] Offset 0xb19cb2: Zip
archive data [!] Offset 0xb5c46f: Zip archive data [!] Offset 0xb8b0fe: Zip archive data [!]
Offset 0xb9b5d6: Zip archive data [!] Offset 0xbb72d8: Zip archive data [!] Offset 0xbc328e:
Zip archive data [!] Offset 0xbcb33a: Zip archive data [!] Offset 0xc7a91c: Zip archive data
[!] Offset 0xc8daeb: Zip archive data [!] Offset 0xc97413: Zip archive data [!] Offset
0xcc7a1f: Zip archive data [!] Offset 0xce265e: Zip archive data [!] Offset 0xced1b5: Zip
archive data [!] Offset 0x16f2e87: JPEG image data [!] Offset 0x19c3d3e: Zip archive data [!]
Offset 0x19c3d9d: Zip archive data [!] Offset 0x1a66f33: JPEG image data [!] Offset 0x1b18bef:
Zip archive data [!] Offset 0x1b18c44: Zip archive data [!] Offset 0x1b1e17d: Zip archive data
[!] Offset 0x1b1e1bc: Zip archive data [!] Offset 0x1b700e4: JPEG image data [!] Offset
0x1bd1b3c: Zip archive data [!] Offset 0x1f98b7c: Zip archive data [!] Offset 0x2216711: JPEG
image data [!] Offset 0x28d08d2: bzip2 compressed data [!] Offset 0x329879e: JPEG image data
[!] Offset 0x35698c9: Zip archive data [!] Offset 0x3780d0c: gzip compressed data [!] Offset
0x38def1e: gzip compressed data [!] Offset 0x399d8e2: Zip archive data [!] Offset 0x3cecec2:
gzip compressed data [!] Offset 0x3e4b0c6: gzip compressed data [!] Offset 0x3f09ac1: Zip
archive data [!] Offset 0x42b59ec: Zip archive data [!] Offset 0x45292e7: Zip archive data [!]
Offset 0x452931c: Zip archive data [!] Offset 0x479cbf0: Zip archive data [!] Offset 0x4eaf722:
bzip2 compressed data [!] Offset 0x5436d53: bzip2 compressed data [!] Offset 0x58bc604: Zip
archive data [!] Offset 0x59c0e05: Zip archive data [!] Offset 0x5af4b04: Zip archive data [!]
Offset 0x6036279: bzip2 compressed data [!] Offset 0x64ecf1c: JPEG image data [!] Offset
0x71f9f9f: bzip2 compressed data [!] Offset 0x7780d7a: bzip2 compressed data [!] Offset
0x7bb6a80: JPEG image data [!] Offset 0x86981ab: Zip archive data [!] Offset 0x86985d6: Zip
archive data [!] Offset 0x89e2ea2: gzip compressed data [!] Offset 0x8b40f27: gzip compressed
data [!] Offset 0x8bff809: Zip archive data [!] Offset 0x8f4e9dd: gzip compressed data [!]
Offset 0x90aca4e: gzip compressed data [!] Offset 0x916b37e: Zip archive data [!] Offset
0x9516e8b: Zip archive data [!] Offset 0x978a46b: Zip archive data [!] Offset 0x978a4a0: Zip
archive data [!] Offset 0x99fda77: Zip archive data [!] Offset 0x9e4fb6a: gzip compressed data
[!] Offset 0xab16c76: gzip compressed data [!] Offset 0xaecd4e8: JPEG image data [!] Offset
0xaf2bf3d: JPEG image data [!] Offset 0xbd105fe: JPEG image data [!] Offset 0xc948216: JPEG
image data [!] Offset 0xcc19493: Zip archive data [!] Offset 0xcc194cc: Zip archive data [!]
Offset 0xcc19512: Zip archive data [!] Offset 0xcc1957f: Zip archive data [!] Offset 0xce031d2:
Zip archive data [!] Offset 0xce36370: Zip archive data [!] Offset 0xce363d5: Zip archive data
[!] Offset 0xd020028: Zip archive data [!] Offset 0xd110211: Zip archive data [!] Offset
0xd110266: Zip archive data [!] Offset 0xd1131c1: Zip archive data [!] Offset 0xd4b1bf6: gzip
compressed data [!] Offset 0xd5327de: gzip compressed data [!] Offset 0xd777846: gzip
compressed data [!] Offset 0xdb480b0: JPEG image data [!] Offset 0xdc6d252: gzip compressed
data [!] Offset 0xded442e: JPEG image data [!] Offset 0xe0476c4: JPEG image data [!] Offset
0xe074887: JPEG image data [!] Offset 0xe2c00ad: JPEG image data [!] Offset 0xe3e7778: bzip2
compressed data [!] Offset 0xe410e83: Zip archive data [!] Offset 0xe410ef8: Zip archive data
[!] Offset 0xe413e53: Zip archive data [!] Offset 0xe5e1efd: bzip2 compressed data [!] Offset
0xe7c8b89: Zip archive data [!] Offset 0xea004a5: bzip2 compress ...[truncated]
```

Tool: CUTTER

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/cutter.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip" --project "29" --run-id 0

```
Starting Cutter (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis
workflow simulator\backend\data\uploads\29_Firmware_120626.zip... No interesting architectures
```

```
found. Scan complete.
```

Tool: GHIDRA

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/ghidra.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip" --project "29" --run-id 0

```
Starting Ghidra (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis
workflow simulator\backend\data\uploads\29_Firmware_120626.zip... No interesting RE artifacts
found. Scan complete.
```

Tool: TRUFFLEHOG

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/trufflehog.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip" --project "29" --run-id 0

```
Starting Secret Scan on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow
simulator\backend\data\uploads\29_Firmware_120626.zip... No obvious secrets found in the raw
binary. Scan complete.
```

Tool: ENTROPY

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/entropy.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip" --project "29" --run-id 0

```
Starting Entropy Analysis on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow
simulator\backend\data\uploads\29_Firmware_120626.zip... Overall Shannon Entropy: 7.9966 Found
1 finding(s). [!] High entropy (8.00) indicates packed or encrypted firmware section Scan
complete.
```

Tool: WIRESHARK

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/wireshark.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip" --project "29" --run-id 0

```
Starting Wireshark (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis
workflow simulator\backend\data\uploads\29_Firmware_120626.zip... No interesting network
packets found. Scan complete.
```

Tool: AFL++

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/afl.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip" --project "29" --run-id 0

```
Starting AFL++ (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip... No obvious fuzzing entry points found. Scan complete.
```

Tool: ANGR

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/angr.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip" --project "29" --run-id 0

```
Starting angr (Lightweight Fallback) on C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip... Found 1 finding(s). [!] Symbolic: High number of branch instructions (4570825) indicates complex control flow suitable for symbolic execution Scan complete.
```

Tool: SCORECARD

Status: SUCCESS | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/scorecard.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip" --project "29" --run-id 0

```
Starting Risk Scorecard Generation on project 29... Found 1 finding(s). [!] RiskScore: Calculated Aggregate Risk Score: 0/100 based on 132 findings Scan complete.
```

Tool: PDF_REPORT

Status: RUNNING | Command: "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\.venv\Scripts\python.exe" tools/pdf_report.py --target "C:\Users\akars\OneDrive\Desktop\Firmware Analysis workflow simulator\backend\data\uploads\29_Firmware_120626.zip" --project "29" --run-id 0